

Revised Property CDM for review

This revised Property CDM keeps the model utility-agnostic and information-first. It separates:

- intrinsic property facts,
- hazard and exposure facts,
- protection and resilience measures,
- external assessment ratings,
- and financial / transaction information.

The structure below reflects the latest decisions:

- resilience fields are grouped by function, not by BRI utility;
- InsurancePremium is removed from RiskAssessment;
- insurance and governing-body ratings use parallel structures;
- BRI-specific outputs are limited to rating metadata, not assessment checklist field names;
- BRIRatingAgent is used instead of BRIRatingBody.

1. PropertyHeader

1.1 Header

- UPRN
- PropertyID
- CatchmentID
- propertyType
- propertyStatus

1.2 Valuation

- PropertyValue
- ValuationDate
- ValuationMethod

1.3 PropertyAttributes

- OccupancyType
- PropertyAreaSqm
- PropertyResi
- NumberOfStoreys
- ConstructionYear
- PropertyPeriod
- CouncilTaxBand
- NumberBedrooms
- NumberBathrooms
- PropertyCondition

1.4 Construction

- ConstructionType
- FoundationType
- FloorLevelMeters
- BasementPresent

1.5 Location

- BuildingNumber
- StreetName
- TownCity
- County
- Postcode
- LocalAuthority
- Country
- Region
- UrbanRuralClassification
- LatitudeDegrees
- LongitudeDegrees

1.6 RiskAssessment

This section holds hazard and exposure facts about the property or site.

- EAFloodZone
- OverallFloodRisk

- FloodRiskType
- GroundLevelMeters
- RiverDistanceMeters

2. ProtectionMeasures

2.1 RiskAssessment

This section holds assessment outcomes and rating metadata, not financial information.

2.1.1 InsuranceBodyRatings

- InsuranceRating
- InsuranceDate
- InsuranceRatingVersion
- InsuranceRatingBody

2.1.2 GoverningBodyRatings

- BRIRating
- BRIDate
- BRIRatingVersion
- BRIRatingAgent

2.2 HazardProfile

This section holds normalized hazard classes used by resilience assessments and downstream aggregators.

- FloodHazardClass
- WindHazardClass
- SeismicHazardClass
- FireHazardClass

2.3 ResilienceMeasures

2.3.1 BuildingAssessment

This section holds resilience checks related to the building fabric, structure, envelope, and engineering robustness.

- SiteExposureAssessed
- OrientationMitigatesWind

- StructureDesignedForHazardWind
- ContinuousLoadPathProvided
- LateralBracingAdequate
- RoofRatedForDesignWind
- RoofEdgeDetailWindResistant
- CladdingRatedForDesignWind
- OpeningsWindResistant
- LargeDoorsReinforced
- RooftopEquipmentAnchored
- FacadeAttachmentsAnchored
- StructuralFireResistanceAdequate
- CompartmentsProvided
- FireStoppingAtPenetrations
- ExternalMaterialsFireResistant
- SiteGeotechnicalAssessed
- StructureMeetsSeismicCode
- SeismicDetailingToStandard
- StructuralRegularityAdequate
- FoundationSuitableForHazard
- NonstructuralComponentsAnchored
- HeavyEquipmentAnchored

2.3.2 SiteAndDrainage

This section holds resilience checks related to site layout, drainage, flood management, and land stability.

- SiteFloodHazardAssessed
- HighRiskZoneAvoidedOrJustified
- FinishedFloorAboveDesignFlood
- OccupiedLevelsElevated
- BasementFloodStrategy
- OnsiteDrainageSizedForDesignStorm
- OverlandFlowPathsMaintained
- PermeableOrRetentionMeasures
- WildfireDefensibleSpace
- WildfireNonCombustiblePerimeter

- HighRiskGroundAvoidedOrMitigated
- LiquefactionMitigationProvided

2.3.3 FloodProtection

This section holds installed flood resilience and flood-protection measures.

- PermanentFloodProofingAtEntries
- DeployableBarriersProvided
- BackflowPreventionInstalled
- ElectricalSystemsAboveFlood
- MechanicalSystemsAboveFlood
- FuelAndHazardousStoresProtected
- FloodGates
- FloodBarriers
- SumpPump
- FloodWarningSystem

2.3.4 FireProtection

This section holds active fire-protection systems.

- AutomaticDetectionInstalled
- SuppressionSystemsInstalled

2.3.5 ContinuityMeasures

This section holds measures that support continuity of operation during or after hazard events.

- BackupPowerInstalled
- BackupPowerProtectedFromHazard
- BackupWaterSupplyProvided
- WaterSystemsProtectedFromHazard
- TelecomRedundancyProvided
- CriticalITProtected
- AccessRouteResilient
- AccessDesignConsidersHazards
- BusinessContinuityPlanInPlace
- EmergencyProceduresTested

3. TransactionHistory

3.1 Purchase

- PurchaseDate
- PurchasePriceGbp

3.2 Rental

- MonthlyRentGbp
- RentalYield

4. Controlled vocabularies and menu candidates

The current schema already uses menu fields extensively, so the revised CDM should continue to use controlled vocabularies where binary yes/no is too limiting.

4.1 Existing menu fields already in the CDM

- propertyType
- propertyStatus
- ValuationMethod
- OccupancyType
- PropertyResi
- PropertyPeriod
- CouncilTaxBand
- PropertyCondition
- ConstructionType
- FoundationType
- Country
- Region
- UrbanRuralClassification
- EAFloodZone
- OverallFloodRisk
- FloodRiskType

4.2 New fields that should likely be menu-based

- InsuranceRating
- InsuranceRatingVersion
- InsuranceRatingBody
- BRIRating
- BRIRatingVersion
- BRIRatingAgent
- FloodHazardClass
- WindHazardClass
- SeismicHazardClass
- FireHazardClass
- BasementFloodStrategy

4.3 Candidate fields that may later evolve from boolean to menu/enum

These are initially practical as booleans, but may later need richer states such as Not assessed, Partial, Meets minimum, Enhanced, Or Verified.

- StructureDesignedForHazardWind
- StructureMeetsSeismicCode
- StructuralFireResistanceAdequate
- OnsiteDrainageSizedForDesignStorm
- ElectricalSystemsAboveFlood
- MechanicalSystemsAboveFlood
- BackupPowerInstalled
- BackupWaterSupplyProvided
- TelecomRedundancyProvided
- EmergencyProceduresTested

5. Review notes

5.1 Clear separation of concerns

- `PropertyHeader.RiskAssessment` = hazard and exposure facts.
- `ProtectionMeasures.RiskAssessment` = assessment and rating outputs.
- `ProtectionMeasures.ResilienceMeasures` = resilience features, checks, and mitigations.
- `TransactionHistory` = commercial / market history.

5.2 Why this works

This keeps the CDM descriptive rather than framework-driven. BRI can consume the hazard classes, resilience checks, and BRI rating metadata later, but the core model remains reusable for insurers, lenders, local authorities, and other reporting frameworks.

5.3 Open design choices for later

- Whether `HazardProfile` should sit under `PropertyHeader.RiskAssessment` or remain under `ProtectionMeasures`.
- Whether `InsuranceBodyRatings` and `GoverningBodyRatings` should be generic enough to support multiple named schemes under each.
- Whether some resilience checks should be split further into `WindProtection`, `SeismicProtection`, and `EnvelopeProtection`.